

Synopsis: Asymmetric Baryogenesis from Electroweak Hysterical Instability in a Cooling Universe

A Standard Model residual mechanism at $t \sim 10^{-11}$ s

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September 2026

Computational note. Proofs, exploratory experiments, and paper writing were assisted by generative AI. Mathematical theorems stated in this paper are formally verified in Lean 4 in the companion specifications, as faithful ports of the TeX arguments at the abstractions recorded there, except results explicitly tagged as *literature hypotheses* (standard theorems cited from the literature and not re-proved here) or *physical inputs* (modeling assumptions outside mathematics).

Abstract

The paper provides a mechanism for asymmetric baryogenesis: cosmological cooling drives the electroweak Hysterical residual through a unique criticality $T_\diamond$; the concurrent epoch is exactly $\mathcal{C} = (T_*, T_\diamond)$, nonempty iff $T_\diamond > T_*$; a CP-odd roll yields $\eta_B \neq 0$ on the selected branch at $t \sim 10^{-11}$ s. No extra inequality $T_\diamond > T_c$ is assumed.

1 Claim

Asymmetric baryogenesis from the Standard Model electroweak residual under cooling—no new force carrier.

2 Forced cooling criticality

Theorem 2.1 (Forced reciprocal criticality under cooling). *For $k_X = \alpha_X T^2$, $k_Y = \alpha_Y T^2$, $g \neq 0$, and $T_\diamond = (g^2/(\alpha_X \alpha_Y))^{1/4}$, $\Delta = \alpha_X \alpha_Y T^4 - g^2$ is positive for $T > T_\diamond$, zero at $T_\diamond$, and negative for $0 < T < T_\diamond$. High- T cooling necessarily begins subcritical and crosses criticality once.*

3 Sharp concurrency

Theorem 3.1 (Concurrent epoch). *$T \in \mathcal{C}$ if and only if $T_* < T < T_\diamond$. Equivalently $\mathcal{C} \neq \emptyset$ iff $T_\diamond > T_*$ (criticality before freeze-out).*

4 Baryogenesis

Theorem 4.1 (Electroweak Hysterical baryogenesis). *If $T_\diamond > T_*$, a CP-odd roll through $\mathcal{C} = (T_*, T_\diamond)$ with portal $YW\widetilde{W}$ yields*

$$\eta_B[Y_+] = -\eta_B[Y_-] \neq 0.$$

Radiation-era timing with lattice edges: $t_\diamond \sim 10^{-11}$ s.

5 Conclusion

Cooling forces unique $T_\diamond$; concurrency is exactly $T_\diamond > T_*$; a CP-odd roll yields asymmetric baryogenesis at $t \sim 10^{-11}$ s.

EW residual + $T: \infty \rightarrow 0 \implies \eta_B[Y_+] = -\eta_B[Y_-] \neq 0$ at the electroweak epoch.